# Large Language Models for Social Science Research

*Session 2. Structured Data with an Agent*

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# Today's Session Overview (9:00–12:00)

**9:00–9:10** Last Week, and What to Keep in Mind (10 min)

**9:10–9:30** How I Actually Work (20 min)

**9:30–9:45** Tokens and Models (15 min)

**9:45–10:25** Block 1. Document and Specify (40 min), then a 15-min break

- Data README, codebook, analysis plan (06)

**10:40–11:15** Block 2. Clean, Merge, Describe (35 min)

- Scripts that check themselves; statistics, plots, first insights (07)

**11:15–11:45** Block 3. Audit and Decide (30 min)

- A fresh reviewer (08), then one step by hand (09)

**11:45–12:00** Debrief and Next Session (15 min)

Slower than last time: every block is *I show* (10 min), *you do*, *we check* (5 min).

Today is structured data. Brought a text corpus? Keep it for Sessions 3 and 4.

# Last Week

*9:00 – 9:10*

What we saw, and what to keep in mind

# Last Week, in One Slide

**The argument.** An AI system now writes economics papers on its own (APE). Generating a paper is almost free; verifying one is not. The bottleneck moved from *doing* to *judging*, and the chain has to end with a human who understands.

**The position.** Delegate the work. Keep the skill to check it, and the responsibility for it. Three rules:

1. You delegate execution, never understanding.
2. Read before you accept.
3. Stop when you no longer know what it is doing.

**The tool.** A folder that remembers: `README.md` (what the project is), `CLAUDE.md` (how the agent behaves), `PROJECT_HISTORY.md` (where we stand). Raw data read-only, prompts versioned, every session ends with a commit.

**The exercise.** Literature notes on your papers, then one note checked against the PDF, looking for the agent's mistake.

# What to Keep in Mind

**This is still an open debate.** How much to delegate, and how much to check, is not settled. Reasonable people draw the line in different places.

**Since last week, I asked many colleagues one question:**

## Where is your limit on review?

What do you read line by line, and what do you let go?

**Today, one answer I find convincing**, from Germain Gauthier (Bocconi). Not *the* answer: a workable one, which you can adapt.

# How I Actually Work

*9:10 – 9:30*

Inspired by Germain Gauthier (Bocconi)

# How I Use AI

- AI is part of pretty much everything I do on my laptop.

**Some use cases**

- A quick overview of papers I am unfamiliar with
- Writing code, organising project folders, updating results in a paper
- Proof-checking that the code, the paper and the slides are consistent
- Discussing ideas and analyses while I work on a project
- Improving the clarity of my writing and presentations

**Pen and paper** when I need to fully understand something (theory, statistics), or for mental clarity (a to-do list, planning ahead).

# How I Do Not Use AI

**The one thing not to do:**

*"Write me a full paper and a full slide set on this vague idea while I go swim in the Limmat."*

- **First, current AIs are terrible at this when left unattended.** It is a very iterative process, and you need to be in the loop.
- **Second, you will present the paper and the results.** Being in the loop is how you understand what you are doing.

The journey is almost as important as the destination in research.

AI can still speed things up, and by a lot. Last week's APE is what happens when nobody is in the loop.

# The Setup: One Project Folder

**Germain's agent is connected to five tools**

| | |
|---|---|
| Dropbox | project files and data |
| GitHub | code history and repository |
| Overleaf | paper and slides |
| HPC cluster | heavy computation |
| Gmail | co-authors' comments |

**The project folder**

```
my-project/
  overleaf/   local copy of the paper
  code/       code repository
  data/       raw and processed data
  notes/      working notes
```

**How it runs**

- The agent runs from the project folder.
- He talks to it in the browser, to see PDFs and figures as they are produced.
- The local copy of the paper and the Overleaf project are always in sync.

The same idea as your folder from last week.
You do not need five tools today: one git folder is enough.

# Principle 1: You Are in Charge

- Agents overproduce, expand the scope, and overreact to details.
- Do not let them run unattended for long (overnight), unless the task is very well defined: for example, monitoring jobs on a cluster and relaunching the ones that fail.
- Stay hands-on: in the loop, correcting course, making the judgement calls.
- Being hands-on is also how you keep understanding what you are doing, and keep a map of it in your head, even when you no longer write the code.

# Principle 2: Accuracy, Not Speed or Cost-Cutting

- Run the models at maximum reasoning. For *everything*.
- A mistake in a paper costs far more than extra tokens or waiting time.
- The difference in intelligence between reasoning levels is large.
- More reasoning does not remove the need for verification.

You asked how to save tokens. This principle says: not here.
At 9:30: what it costs, and where you can save without breaking it.

# A Typical Loop

1. **Specify the task in writing.** Sample, equation, estimator, clustering, expected output: as in a detailed pre-analysis plan. Germain writes it in Overleaf; you, in `notes/analysis_plan.md`.

2. **A first agent implements it.** It flags ambiguities *before* coding: there is high value in being on the same page.

3. **Other agents review it in a fresh context.** They read the plan, the code and the outputs, and list the discrepancies.

4. **You decide what to revise or accept.** A reviewer asked to find problems always finds something. Resolve, rerun, repeat until satisfied.

Not exotic: Anthropic's own guide to Claude Code recommends the same steps.

# The Typical Failure: Silent Decisions

A typical failure mode is the AI making consequential decisions without asking. For example, dropping some observations.

**On data, it looks like this**

- rows lost in a merge (inner join instead of left join)
- a missing code (−9, 99) read as a value
- a duplicate removed
- an identifier read as a number (`01001` becomes `1001`)

**Two defences**

- Every script prints the number of rows before and after each step. A change the plan does not explain stops the agent.
- I always ask the reviewing agents to check for this too.

Last week's question: how would you know a merge dropped a tenth of your observations?

# Two Phases: Prototyping, Then Sharpening

**Prototyping phase:**

- It is an exploratory phase.
- I do not review the code or the prose.
- I treat these outputs as probable but provisional.

**Sharpening phase:**

- If I think there is something there, I get involved in the writing and coding process.
- I audit the code, paying special attention to the number of rows in the dataset and results that "look weird". I have many independent agents check the code.
- I rewrite the paper and work with AI to iterate on the visuals.
- I take responsibility for the results and claims.

# Communication with Co-authors

Keep three distinctions explicit.

| | |
|---|---|
| **Origin** | what the AI produced; what I produced |
| **Status** | what is exploratory; what is polished |
| **Responsibility** | what I have not checked; what I am ready to stand behind |

Your co-authors will not look at exploratory and polished results in the same way, and will not make the same comments. AI use and understanding vary across co-authors: be transparent, and take responsibility for your work.

# Tokens and Models

*9:30 – 9:45*

What Principle 2 costs, and where you can save

# What Costs Tokens

- **Every message re-sends the whole conversation** to the model: the files it read, its outputs, all previous turns. Turn 40 re-reads turns 1 to 39.
- **The cache discounts what was already sent**, while it is warm: about an hour on a subscription, five minutes with an API key. Back from lunch, the first message pays full price.
- **Effort multiplies tokens**: more files read, more checks, more thinking.
- **A bigger model costs more per turn.**
- **Subagents** spend tokens while you are not typing.

Subscription: tokens are quota. API key: tokens are money.
`/usage` shows where yours went.

# Model and Effort: Two Dials

**Model: how capable.** A bigger model understands more: unfamiliar libraries, ambiguous instructions, subtle bugs. It costs more per token.

**Effort: how thorough.** The same model works longer: it reads more files, thinks more, checks its own work before answering. It costs more tokens.

**When the answer is wrong, ask which one it is.**

- It *did not know*: it misread an ambiguous variable, or did not know how a package behaves. Change the **model**.
- It *did not try*: it skipped a file, did not rerun the code, did not check. Raise the **effort**.

`/model` to see and change the model. `/effort` for the effort: `low`, `medium`, `high`, `xhigh`, `max`.

# Which Model, Which Effort, for Which Task

| **Task** | **Model** | **Effort** | **Why** |
|---|---|---|---|
| Rename, reformat a table, commit message | Sonnet | `low` | you check each result anyway |
| Prototyping: exploratory scripts, quick plots | Sonnet | `medium` | provisional by design |
| Writing: the paper, memos, the analysis plan | Fable | `high` | long, ambiguous tasks |
| Code that enters the paper: cleaning, merging, estimation | Opus | `high` | errors hide in cases that are easy to overlook |
| Independent review | Opus | `xhigh` | the last check before yours |

Default in Claude Code: Opus 5.5 at `medium`. Fable is the most expensive per token: select it with `/model fable`, and keep it for what needs it.

# Save on Context, Not on Reasoning

- **One task, one session.** `/clear` between unrelated tasks costs nothing and loses nothing: the project's memory is in the folder (`PROJECT_HISTORY.md`), not in the chat.
- **Watch the first minute.** `Esc` as soon as it drifts; `/rewind` to go back. After two failed corrections: `/clear` and one better prompt, not a third correction.
- **Never paste data.** The agent writes a script that prints summaries (types, counts, missing values); only the summaries enter the context. It also protects confidential data.
- **Point, do not paste.** `@code/03_merge.py`, not the file's content. PDFs: only the pages you need.
- **A short** `CLAUDE.md`: under about 200 lines. Long procedures go in `prompts/`.
- **A long task:** `/compact` with a focus ("keep the merge decisions"). Back after more than an hour: a fresh session costs less.

Save on context. Never on reasoning for what enters the paper.

# Your Project, Three Blocks

*9:45 – 11:45*

I show, you do, we check

# The Example Project: One Step Further

**In Session 1, at the end of the demo**, the agent downloaded five presidential elections (2002–2022) and cleaned them into one table of municipalities. Its checks passed.

But it took six decisions alone: the scope, the overseas codes, Paris, the far-right candidate, Zemmour, blank ballots. And municipalities merged between elections: the same code does not always mean the same territory.

**Today's question**

Did the far-right vote grow more between 2002 and 2022
where the population declined?

**Three sources, three geographies**

- The five elections: each in the municipal geography of its year
- INSEE historical series, censuses 1968–2015: the geography of 2017
- INSEE history of municipalities: every merger, split and change of code

# What You Leave With at Noon

In your folder:

- `data/raw/README.md` and `notes/codebook.md`: where each file comes from, what each variable means
- `notes/analysis_plan.md`, in your words, with its open questions settled
- numbered scripts in `code/` that print their row counts and check themselves, and `code/run_all.py`
- a cleaned dataset in `data/processed/`, summary statistics, two figures, and `notes/insights.md`
- a review written by a fresh session, and one step you audited by hand
- an entry in `PROJECT_HISTORY.md` and a commit

**And one thing you will have seen:** a decision the agent took on your data without asking. Finding it is Block 3. It is the only step you cannot skip.

# Block 1. Document and Specify (9:45–10:25)

**I show (10 min).** The example project. Prompt 06: a script inspects the three sources without changing them. The agent completes the data README, writes the codebook, interviews me, and writes the plan.

**You do (25 min)**

1. Your data in `data/raw/`. No data? The example pack, in the course repository (`sessions/02/`).
2. Copy prompts 05 to 09 and `notes/templates/` from the course repository (`template/`) into your folder. A new session. Prompt 06.
3. Answer its questions. What you do not know yet goes to "Open questions". It asks; it does not decide.
4. Read the codebook and the plan before you say "ok".

Survey data: the codebook also records the weights, the filter questions, and the codes for refusal, don't know and not asked.

**We check (5 min).** Two codebooks on screen. A variable whose missing-value codes nobody documented? A merge with no rule for the rows that do not match?

# Scripts, Not Chats

- The agent writes code into the project. It does not answer in the chat: a number in a chat is a claim nobody can rerun.
- Every table and figure has a script behind it. `python3 code/run_all.py` rebuilds everything from `data/raw/`.
- Coding assistance, three uses:
  - Writing: one script per step, each saying what it reads, writes and drops.
  - Debugging: paste the error, and ask for the cause before the fix.
  - Documenting: a docstring at the top of each script, the codebook for the variables.
- Reading the code stays your job. Ask the agent to explain it.

Developers learning a new library with an AI assistant scored 50% on a quiz, against 67% for those coding by hand. Those who asked the AI for explanations scored about as well as by hand (Shen and Tamkin 2026).

# Checks to Request

Silent errors: bad merges, dropped observations, miscoded missing values, mislabelled variables. Three checks catch most of them. Ask for them in the code, not in the chat.

**Row counts**, before and after every step

```
merge: 1,000 rows -> 980; unmatched: 20 left, 5 right
```

**Assertions**, which stop the script when they fail

```
assert df["code_commune"].is_unique
assert df["code_commune"].str.len().eq(5).all()
assert df["share_fnrn"].between(0, 1).all()
assert merged["votes"].sum() == raw["votes"].sum()
```

**Cross-tabulations**, after every recode and every merge

```
pd.crosstab(df["q12_raw"], df["q12_clean"], dropna=False)
pd.crosstab(merged["_merge"], merged["region"])
```

# Block 2. Clean, Merge, Describe (10:40–11:15)

**I show (10 min).** `/clear`, then prompt 07. The agent lists what the plan leaves open; I answer. It writes the scripts, with their checks, then the summary statistics, two figures and its first insights. We read the row counts first, the figures last.

**You do (20 min)**

1. `/clear`: the plan is in the folder, not in the chat. Prompt 07.
2. Answer its list of ambiguities before it writes any code.
3. Read in this order: the row counts, the assertions, the cross-tabulations, then the figures and the insights.
4. Rows that change without the plan saying so: paste the step in the shared document.

**We check (5 min).** The shared document: who lost rows, at which step, and was it in the plan?

# When to Trust, When to Intervene

**Trust a step** when a check exists for it, it passed, and you can rerun it.

**Intervene** when

- a number moved and nobody can say why
- a match is right on the code and wrong on the territory, the wave or the period
- a label says one thing and the code computes another
- the result looks too clean

**Audit one step by hand** (prompt 09)

1. Choose the riskiest step: a merge, a recode, a filter.
2. Read its code. Ask the agent to explain each line.
3. Follow three units from the raw file to the final table: one ordinary, one edge case, one dropped or unmatched.
4. Read their values in the raw file yourself, and recompute one number by hand.

# Block 3. Audit and Decide (11:15–11:45)

**I show (10 min).** I close the session and open a fresh one. Prompt 08: a reviewer that did not see the code being written. Then prompt 09: I audit one step by hand, three municipalities that became one.

**You do (15 min)**

1. Close the session. Open a new one. Prompt 08. It runs for several minutes: do not wait.
2. Prompt 09 on that step, in a second session. Read three units in the raw data yourself. Not optional.
3. For each point of the review and of your audit: fix it, or accept it and write down why. Rerun `code/run_all.py`.
4. Prompt 05: the entry in `PROJECT_HISTORY.md`, and the commit.

**We check.** That is the debrief.

# Debrief

**Three screens.** A codebook. A row-count trail. A step audited by hand, with what it found.

**The decisions the agent took alone today.** Read from the shared document. The list is the lesson.

**One question.** The table you would have sent your supervisor an hour ago: which status, exploratory or polished, would it have carried?

**Before you leave.** Ask the agent to copy the section "Row counts and silent decisions" of `template/CLAUDE.md` into yours. Then add one line of your own that would have caught today's error.

# Next Session: 4 November 2026

**Session 3: Building Datasets from APIs, and Classic Text-as-Data**

- Collecting structured and unstructured data through APIs, with the agent writing the collector
- Information extraction without an LLM: dictionaries, embeddings
- Bring one source you would like to collect from: an API, a website, a set of documents
- Keep working in your folder. End every session with prompt 05.

## Carry-over question

An API returns 10,000 documents and the agent reports that the collection is complete. How would you know?

# References


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